CS24305 — THEORY (3.0 CR) | MODULE II NOTES | BIT MESRA | NEP SCHEME

Module 2: Data Link Layer & ARQ

Framing (Bit Stuffing), CRC-32 Division, Hamming Code, Stop-and-Wait, Go-Back-N & Selective Repeat

Module II: Data Link Layer, Error Control & Flow Control ARQs

1. Data Link Layer Design Issues & Service Models to Network Layer
2. Framing Mechanisms: Byte Count, Byte Stuffing & Bit Stuffing with Flag `01111110`
3. Error Detection: Simple Parity, 2D Parity & Internet Checksum Algorithm
4. Cyclic Redundancy Check (CRC-32): Polynomial Division & Full Worked Traces
5. Error Correction: Hamming Code (n, k) Derivations & Parity Check Matrices
6. Flow Control Mechanics: Stop-and-Wait Protocol & Channel Efficiency (η)
7. Sliding Window Concepts: Bandwidth-Delay Product & Piggybacking
8. Go-Back-N (GBN) ARQ Protocol: Sender/Receiver Windows ($W_s \leq 2^m - 1$)
9. Selective Repeat (SR) ARQ Protocol: Independent ACKs & Windows ($W_s \leq 2^{m-1}$)
10. Channel Utilization & Efficiency Proofs: $\eta = \frac{W}{1+2a}$ where $a = \frac{T_p}{T_t}$
11. High-Level Data Link Control (HDLC) Frame Formats & PPP Architecture
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 1 & 2: Data Link Framing Mechanisms

The Data Link Layer converts the raw bit stream delivered by the Physical Layer into discrete, manageable packets called **Frames**:

Framing Method	Delimiting Mechanism & Bit/Byte Manipulation	Key Failure Modes & Mitigations
1. Character / Byte Count	Header field contains an integer specifying total number of bytes in frame.	Fatal synchronization loss: if a transmission error alters the count byte, all subsequent frames are misframed.
2. Byte Stuffing (Character-Oriented)	Frames bounded by `FLAG` bytes (`0x7E`). Any occurrence of `FLAG` or escape byte `ESC` (`0x7D`) in payload is prefixed with an `ESC` byte.	Tied to specific 8-bit character encodings (ASCII).
3. Bit Stuffing (Bit-Oriented — HDLC)	Frames delimited by flag pattern `01111110` (6 consecutive `1`s). Whenever sender observes 5 consecutive `1`s in data payload, it unconditionally stuffs a `0` bit.	Completely code-transparent; handles arbitrary arbitrary binary streams.

Topic 4: Cyclic Redundancy Check (CRC) Polynomial Division

CRC Transmission Invariants: Let data bit sequence be represented by polynomial $D(x)$ of length k, and Generator Polynomial be $G(x)$ of degree r. 1. Append r zero bits to $D(x) \implies D(x) \cdot x^r$. 2. Divide $D(x) \cdot x^r$ by $G(x)$ using modulo-2 arithmetic (XOR binary division). 3. The remainder $R(x)$ of length r is the CRC Checksum. Transmitted codeword $T(x) = D(x) \cdot x^r \oplus R(x)$.

🏛 Step-by-Step Solved Problem: CRC-4 Calculation on Data `1101011011` with Generator $G(x) = x^4 + x + 1$

Step 1: Convert Generator to Binary String: $G(x) = 1 \cdot x^4 + 0 \cdot x^3 + 0 \cdot x^2 + 1 \cdot x^1 + 1 \cdot x^0 \implies \mathbf{10011}$ (Degree $r = 4$).

Step 2: Append $r = 4$ zeros to Data ($D = 1101011011$): Augmented Dividend: $\mathbf{11010110110000}$.

Step 3: Modulo-2 Polynomial XOR Division:

```
            1100001010  (Quotient)
10011 ) 11010110110000
        10011
        _____
         10011
         10011
         _____
          000010110
              10011
              _____
               010100
                10011
                _____
                 01110  ⟵  Remainder R = 1110
```

Step 4: Transmitted Codeword: $11010110110000 \oplus 1110 = \mathbf{11010110111110}$.

Topic 6 – 10: Sliding Window Flow Control ARQ Protocols

Protocol	Sender Window (W_s)	Receiver Window (W_r)	Channel Efficiency (η)
1. Stop-and-Wait	$W_s = 1$	$W_r = 1$	$\eta = \frac{T_t}{T_t + 2T_p} = \frac{1}{1+2a}$
2. Go-Back-N (GBN)	$W_s = 2^m - 1$	$W_r = 1$ (Cumulative ACK)	$\eta = \frac{W_s}{1+2a} \quad (\text{if } W_s < 1+2a, \text{ else } 100\%)$
3. Selective Repeat (SR)	$W_s = 2^{m-1}$	$W_r = 2^{m-1}$ (Independent ACK)	$\eta = \frac{W_s}{1+2a}$

Where m = number of sequence number bits, $T_t = \frac{L}{B}$ = transmission time, $T_p = \frac{d}{v}$ = propagation delay, $a = \frac{T_p}{T_t}$.

🏛️ Top BIT Mesra Exam Questions & Answers (Module II)

Q1. Why must the maximum sender window size in Selective Repeat ARQ satisfy $W_s \leq 2^{m-1}$? (8 Marks)

If $W_s + W_r > 2^m$, an ambiguous overlap occurs between the old sequence number window and the new sequence number window.
Proof: Suppose $m = 2$ (sequence numbers $0, 1, 2, 3$) and $W_s = W_r = 3$. Sender transmits frames $0, 1, 2$. Receiver receives all 3 and shifts window to $[3, 0, 1]$, sending ACKs $0, 1, 2$. If all ACKs are lost in transit, sender times out and retransmits frame 0. The receiver, expecting new frame 0 in its current window $[3, 0, 1]$, will accept the duplicate old frame 0 as new data, causing undetected duplicate data corruption! To prevent window overlap, $W_s + W_r \leq 2^m \implies W_s \leq 2^{m-1}$.